



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Team 45: LES Parts Database UI

Bi-Weekly Update 5

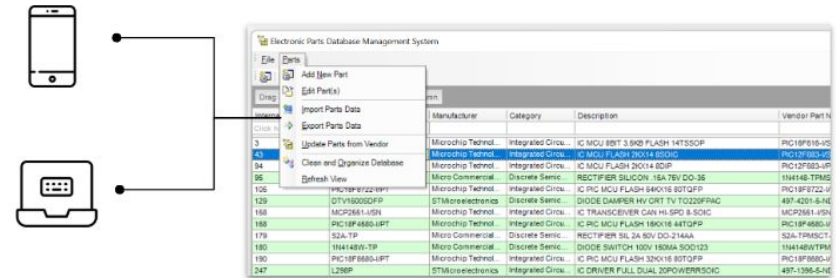
Anthony Beneventi, Jamaal Bloom, Jaswin Jabbal

Sponsor: John Lusher

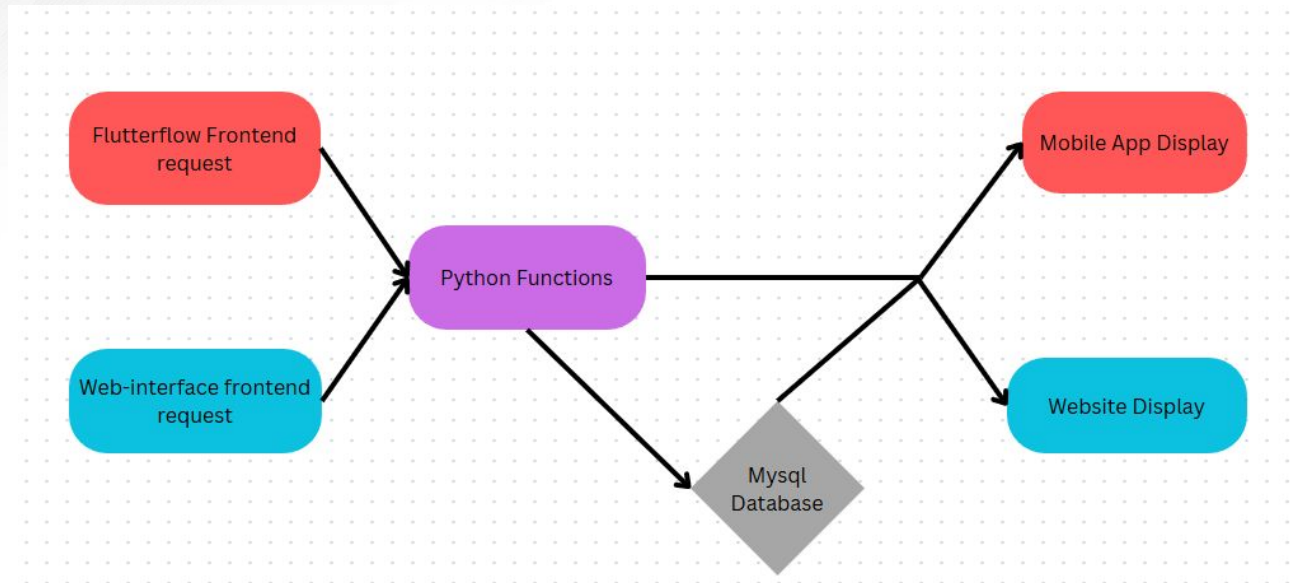
TA: Sabyasachi Gupta

Project Summary

- The current database requires manually updating parts and limits multiple users working at the same time.
- Develop a web and mobile application that will allow for the auto population and filtering of part information

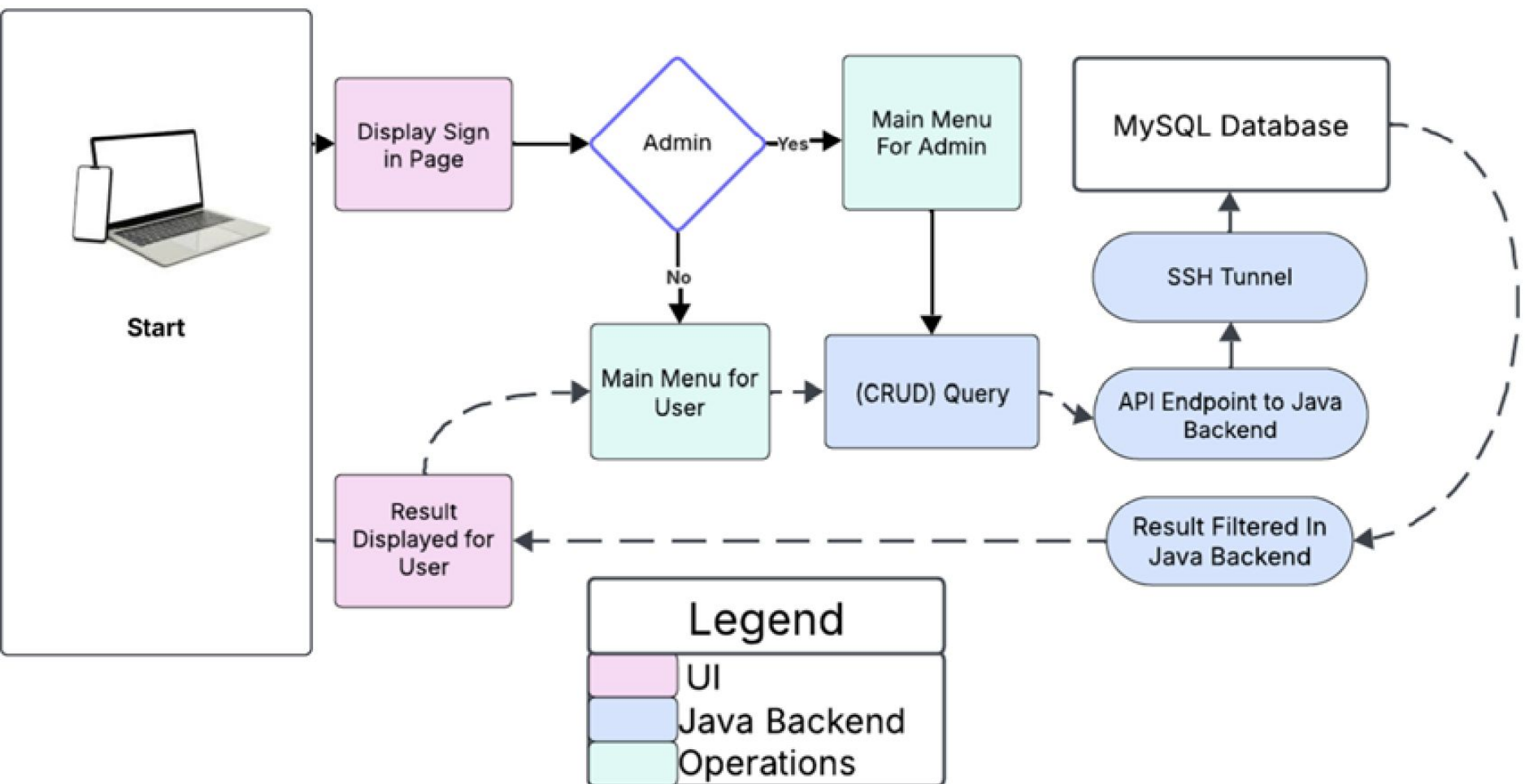


Project/Subsystem Overview



Team: 
 Jamaal: 
 Anthony: 
 Jaswin: 

Integrated System Diagram





Project Timeline

Subsystem Designs and Testing (completed 1/27)	Start work on a singular database (completed 2/20)	Connect backend API to frontend (completed 2/27)	General Integration (completed 3/6)	System Test - CRUD functionality (completed 3/17)	Validation + Deployment (to complete by 3/31)	Demo and Report (to complete by 4/21)
---	---	---	--	--	--	--



Web Application UI

Jaswin Jabbal

Accomplishments since Update 4 40 hrs of effort

- API integration validation
Completed testing/validation with Jamaal's API functions
- Local deployment w/ Docker
app.py can be built + run via Docker containers - ideal for Flask

Ongoing progress/problems and plans until the final presentation

- Final deploy to EC2
Encountering difficulties with SSH integration into Docker - solution: SSH tunnel as system service
- Integrate Google authentication?

← → 127.0.0.1:5000/login

Lusher Engineering Services Login

Login

Username

Password

Login as:

User

Login

New user? [Register here](#)

© 2025 Lusher Engineering Services - Electronic Parts Catalog

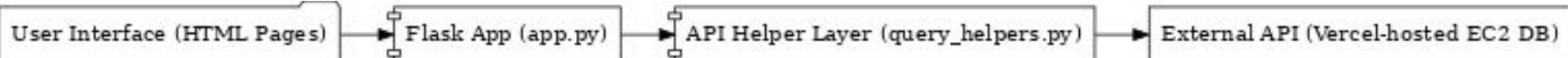
Web Application UI

Web-based inventory system for managing electronic parts, powered by:

Flask (Python) | MySQL (remote) | Docker (build/run) | SSH Tunnel (home page) | Role-based Login System (Admin/User) | Bootstrap Frontend

Features

- Filter, Sort parts
- Add, Update, Tag, Delete parts (Admin Only)
- User registration, login, and session management
- Remote MySQL database access through a secure SSH tunnel
 - Dockerized application (coming soon!)
- Clean & modern UI (Bootstrap + animations)





Python Backend

Jamaal Bloom

Accomplishments since Update 4 25 hrs of effort	Ongoing progress/problems and plans until the next presentation
● Collaborated with teammates to ensure same database and table are being used. ● Verified that requests come through from both frontends	● Continue testing and validation ● Continue refining functions



Python Backend

```
def Add():
    url = 'https://404-app-integration.vercel.app/Add'
    headers = {"Authorization": "Bearer iwtZuWF0U3wmxs4U4Jfm8Z6b"}
    data = {'partnumber': '1N4448XTPMCT-ND'} #Partnumber to be added, Currently needs to be supplier part number and specifac
    response = requests.post(url, data=data, headers=headers)
    return(response.text)

def Tag():
    url = 'https://404-app-integration.vercel.app/Tag'
    headers = {"Authorization": "Bearer iwtZuWF0U3wmxs4U4Jfm8Z6b"}
    data = {'partnumber': '1N4448XTPMCT-ND', #Partnumber where tag is to be added
           'tag': "Test 4"} #Tag to be added
    response = requests.post(url, data=data, headers=headers)
    return(response.text)

def Update():
    url = 'https://404-app-integration.vercel.app/Update'
    headers = {"Authorization": "Bearer iwtZuWF0U3wmxs4U4Jfm8Z6b"}
    data = {'lower': '0', #lower bound
           'upper': '10'} #upper bound, Recommended don't do more then like 10 as it may time out
    response = requests.post(url, data=data, headers=headers)
    return(response.text)
```

ID	Internal PN	Part Description	Manufacturer	Manufacturer Part Number	Supplier 1	Supplier Part Number 1	Part Category
1	1	IC PREC OP-AMP UNIVRSL SNGL 8DIP	Linear Technology	LT1008CN8	Digi-Key	LT1008CN8-ND	Integrated Circuits (ICs)
3	3	IC MCU 8BIT 3.5KB FLASH 14TSSOP	Microchip Technology	PIC16F616-I/ST	Digi-Key	PIC16F616-I/ST-ND	Integrated Circuits (ICs)
4	4	RES SMD 10K OHM 1% 1/8W 0805	YAGEO	9C08052A1002FKHFT	Digi-key	311-10.0KCCT-ND	Resistors
6	6	MOSFET 2N-CH 20V 8A 8SOIC	onsemi	FDS6894A	Digi-key	FDS6894A-ND	Discrete Semiconductor F
7	7	CLIP BATTERY AA PC MNT PAIR	Keystone Electronics	92	Digi-Key	92K-ND	Battery Products
8	8	MOSFET 2N-CH 20V 9.4A 8SOIC	onsemi	FDS6898A	Digikey	FDS6898ACT-ND	Discrete Semiconductor F
9	9	IC OPAMP GP R-R 1MHZ SOT23-8	Texas Instruments	OPA2348AIDCNT	Digi-Key	296-13008-1-ND	Integrated Circuits (ICs)
11	11	IC REG BUCK ADJ 1.2A 10HVSSOP	Texas Instruments	TPS62040DGQR	Digikey	296-15219-1-ND	Integrated Circuits (ICs)
12	12	FIXED IND 6.2UH 1.8A 45 MOHM SMD	Sumida America Components Inc.	CDRH5D28-6R2NC	Digi-key	CDRH5D28-6R2NC-ND	Inductors, Coils, Chokes



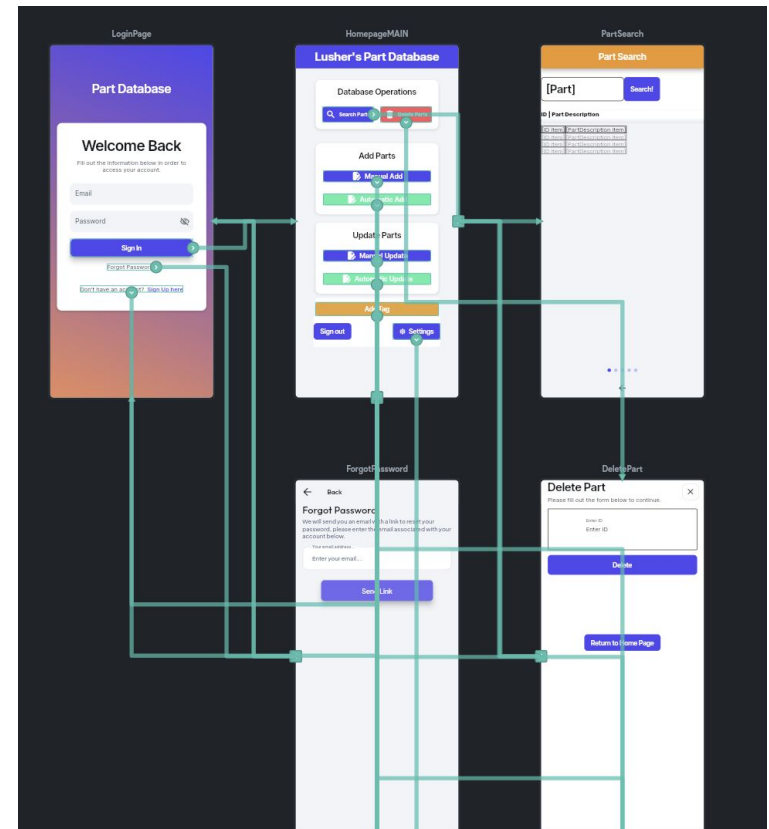
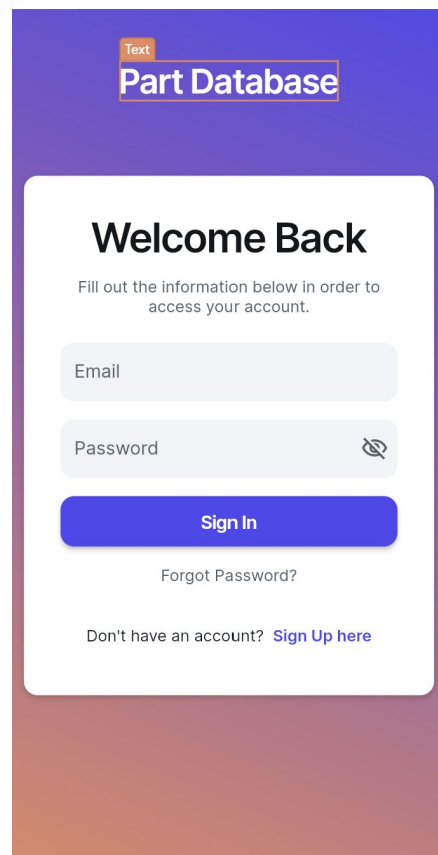
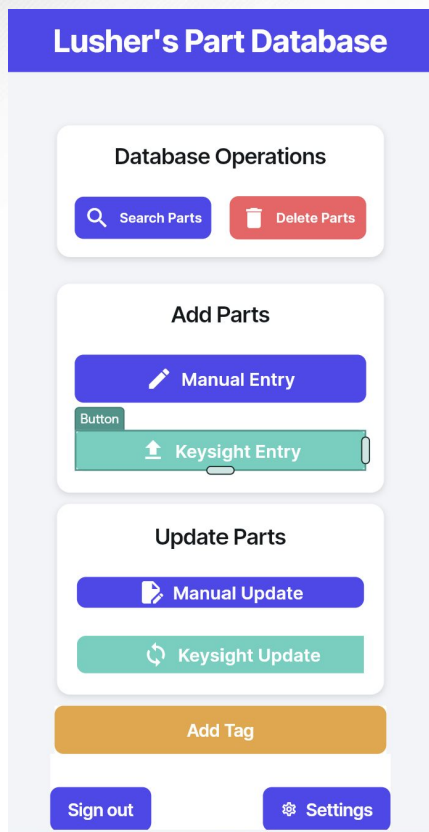
IOS Application

Anthony Beneventi

Accomplishments since Update 4 35+ hrs of effort	Ongoing progress/problems and plans until the next presentation
• Tested functions, limits/ranges • Updated Jamaal's functions for integration • Multiple user functionality • Improved error handling when dealing with irregular input	• Improve UI & Code Appearance • Final Improvements • Deployment

IOS Subsystem

<https://www.youtube.com/watch?v=0rWJT-cg5A4>



Execution Plan

Task Assignments	Owner	1/13/25	1/20/25	1/27/25	2/3/25	2/10/25	2/17/25	2/24/25	3/3/25	3/10/25	3/17/25	3/24/25	3/31/25	4/7/25	4/14/25	4/21/25	4/28/25	Due Date		Key	
Finish Subsystems	Team																			Planned	
Status Update #1	Team																	1/27/25		In Progress	
Convert code to callable API	Jamaal																			Complete	
New models based on DB requirement	Jaswin																			Behind Schedule	
Fix Update/Insert	Anthony																				
Status Update #2	Team																	2/10/25			
Connect with online database	Jamaal																				
Routes based on new models	Jaswin																				
Integrate Database	Anthony																				
Status Update #3	Team																	2/24/25			
Format frontend and backend requests	Jamaal																				
Data Consistency	Jaswin																				
Consolidate App	Anthony																				
Finish general integration	Team																				
Status Update #4	Team																	3/17/25			
Implement failsafes	Jamaal																				
Deployment	Jaswin																				
System validation	Anthony																				
Staus Update #5	Team																	3/31/25			
Finalize code	Jamaal																				
Deployment	Jaswin																				
Optimize code	Anthony																				
Final Presentation	Team																	4/14/25			
Showcase/ Demo	Team																	4/24/25			



Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.4	Database Sync	The web and mobile applications uses the same database and changes visible from both	Attempt changes from multiple devices and ensure consistency	TESTED	Jamaal
3.2.1.1	API Correctness	The API pulls correct information from the supplier	Compare recieved information with suppliers website	TESTED	Jamaal
3.2.6.2	Request Correctness	The backend correctly recieved and handles the frontend requests	Send frontend request from both applications and examine results	TESTED	Jamaal
3.2.1.3	Multiple Requests	Application can handle multiple requests at the same time	Send multiple requests	TESTED	Jamaal
3.2.2.1	Model Validation	All DB models align correctly with the updated schema used by Jamaal	Compare Flask SQLAlchemy models against the latest database schema	TESTED	Jaswin
3.2.4.1	Route Functionality	Form submission routes to the correct display without errors	Manually fill out and submit forms in the application, verifying data correctness	TESTED	Jaswin
3.2.4.2	Frontend Data Consistency	Submitted data is correctly displayed on a corresponding UI template	Manual validation by cross-checking displayed records with DB entries	TESTED	Jaswin
3.2.6.1	Deployment Environment	All required dependencies are correctly installed and configured on the end machine	Validate with Docker, verify installation logs, ensure WGSi server	Untested	Jaswin
3.2.1.2	CRUD Operation Test	Pulling apart information from the MySQL Database	Access a local MySQL Server with dummy datta	TESTED	Anthony
3.2.1.4	Updated Database	The API Correctly updates the entire database	Checking previous and updated dababases to ensure completion	TESTED	Anthony
3.2.1.5	Request Correctness	Having all requested components, such as User authority	Giving specific permission to individual User admins	TESTED	Anthony
3.2.1.3	Application Deployed	Uploaded on the cloud and ready for use	Able to be sent for beta testing	TESTED	Anthony



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 45: Anthony Beneventi, Jamaal Bloom, and Jaswin Jabbal

Questions?